**CS502 - Fundamentals of Algorithms (Quiz No. 3)**

**Question # 7 of 10 ( Start time: 04:38:51 PM, 25 June 2025 )**

What is the value of k in the given equation?

$m[1, 5] = m[1, 3] + m[4, 5] + p0 \cdot p3 \cdot p5$

**Select the correct option**

| | |
|---|---|
| ○ | 4 |
| ● | 3 |
| ○ | 5 |
| ○ | 1 |

**CS502 - Fundamentals of Algorithms (Quiz No. 3)**

**Question # 5 of 10 ( Start time: 04:37:54 PM, 25 June 2025 )**

Huffman algorithm generates an optimum _____ code.

**Select the correct option**

| | |
|---|---|
| ○ | Postfix |
| ○ | Infix |
| ○ | None of the given options |
| ◉ | Prefix |

**CS502 - Fundamentals of Algorithms (Quiz No. 3)**

**Question # 3 of 10 ( Start time: 04:37:23 PM, 25 June 2025 )**

Huffman algorithm produces the _____ prefix code tree.

**Select the correct option**

| | |
|---|---|
| ○ | Worst |
| ○ | Best |
| ○ | Better |
| ◉ | Optimal |

**CS502 - Fundamentals of Algorithms (Quiz No. 3)**

**Question # 9 of 10 ( Start time: 04:39:23 PM, 25 June 2025 )**

The Knapsack problem belongs to the domain of _____ problems.

**Select the correct option**

| | |
|---|---|
| ○ | Searching |
| ◉ | Optimization |
| ○ | Linear solution |
| ○ | Sorting |

**CS502 - Fundamentals of Algorithms (Quiz No. 3)**

**Question # 8 of 10 ( Start time: 04:39:07 PM, 25 June 2025 )**

In Activity scheduling algorithm, the width of a rectangle _____

**Select the correct option**

| | |
|---|---|
| ◉ | Indicates the duration of an activity |
| ○ | Should be maximized |
| ○ | Is always ignored |
| ○ | Directs towards recursion |

**CS502 - Fundamentals of Algorithms (Quiz No. 3)**

**Question # 2 of 10 ( Start time: 04:37:08 PM, 25 June 2025 )**

Those problems in which Greedy finds good, but not always best is called a greedy_____.

**Select the correct option**

| | |
|---|---|
| ○ | Solution |
| ○ | Algorithm |
| ◉ | Heuristic |
| ○ | Result |

**CS502 - Fundamentals of Algorithms (Quiz No. 3)**

**Question # 1 of 10 ( Start time: 04:36:54 PM, 25 June 2025 )**

Chain matrix multiplication problem can be solved through _____ strategy.

**Select the correct option**

| | |
|---|---|
| ○ | greedy |
| ○ | divide and conquer |
| ○ | Implementing arrays |
| ◉ | dynamic programming |

**CS502 - Fundamentals of Algorithms (Quiz No. 3)**

**Question # 4 of 10 ( Start time: 04:37:38 PM, 25 June 2025 )**

Suppose you are given infinite coins of 1, 2, 3, and 4. Select the ways of the minimum number of coins that required to achieve a sum of 6:

**Select the correct option**

| | |
|---|---|
| ○ | 1 |
| ○ | 3 |
| ○ | 4 |
| ◉ | 2 |

**CS502 - Fundamentals of Algorithms (Quiz No. 3)**

**Question # 10 of 10 ( Start time: 04:39:36 PM, 25 June 2025 )**

We can't utilize a divide-and-conquer strategy for Chain Matrix Multiplication because _____.

**Select the correct option**

| | |
|---|---|
| ○ | We use divide and conquer for sorting only |
| ○ | We can easily perform it in linear time |
| ○ | Size of data is not given |
| ◉ | We do not know the optimum k |

**CS502 - Fundamentals of Algorithms (Quiz No. 3)**

**Question # 6 of 10 ( Start time: 04:38:09 PM, 25 June 2025 )**

Suppose we have 4 matrices A, B , C, D . What is correct expansion of m[1,2] in chain matrix multiplication?

**Select the correct option**

| | |
|---|---|
| ○ | $m[1, 2] = m[1, 1] + m[1, 2] + p0 \cdot p1 \cdot p2$ |
| ○ | $m[1, 2] = m[1, 1] + m[2, 2] + p0 \cdot p1 \cdot p3$ |
| ○ | $m[1, 2] = m[1, 2] + m[2, 2] + p0 \cdot p1 \cdot p2$ |
| ● | $m[1, 2] = m[1, 1] + m[2, 2] + p0 \cdot p1 \cdot p2$ |